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Governing OpenAI (A)

In late November 2023, OpenAI's new board of directors took stock of the situation. The company, which sought to develop artificial general intelligence (AGI)—computer systems with capabilities exceeding human abilities—was looking to regain its footing after a chaotic leadership and governance crisis that played out a week earlier. The previous board had stunned observers by firing CEO Sam Altman and removing him from the board for unspecified reasons. Days later, Altman was back as CEO, directors resigned, and a new three-person board formed with Bret Taylor, Larry Summers, and Adam D'Angelo, the only continuing director. (See **Exhibit 1** for directors as of November 29, 2023.)

The new board faced urgent issues—around governance, how to build out the board, AI ethics and safety, and their relationship with a CEO one of them had helped fire. Their quandary was complicated by OpenAI's unique mission to create AGI to benefit all of humanity and by its unusual structure as a non-profit controlling a for-profit. The AI it sought to develop had the potential to be world-changing—for better, or for worse—and the world was watching closely as the board sought a path forward.

OpenAI

OpenAI, Inc. was founded in 2015 as a non-profit registered in Delaware and based in San Francisco. By 2019, its board was the governing body for a group of entities, including for-profit subsidiary OpenAI Global, LLC for which most of its nearly 800 employees worked. The group, referred to simply as "OpenAI," was known for ChatGPT, an artificial intelligence (AI) chatbot. The non-profit disclosed revenue of just \$44,485 in 2022.¹ Although the for-profit's financials were not disclosed, sources put its annualized revenue run rate by October 2023 at \$1.3 billion with sales from ChatGPT subscriptions and payments from developers and enterprises for access to its technology.² OpenAI was said to be in talks with private investors to value it as highly as \$90 billion.³

Management

As of December 2023, OpenAI management included CEO Sam Altman, President Greg Brockman, CTO Mira Murati, and COO Brad Lightcap. Altman, 38, dropped out of Stanford in 2005 to run Loopt, a social application for sharing GPS location. Part of the first class at start-up incubator Y Combinator—which would become famous for backing billion-dollar firms like Airbnb, Stripe, and Dropbox—Loopt

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raised capital from top VCs like Sequoia. However, it never took off and was sold at a loss in 2012.⁴ Still, Altman left a strong impression. Y Combinator co-founder Paul Graham said of him in 2008: “You could parachute him into an island full of cannibals and come back in 5 years and he’d be the king.”⁵

After Loopt, Altman started a VC fund and worked part-time at Y Combinator. He had a reputation as “a well-connected wunderkind.” Explained one founder who said he’d thought a lot about what made Altman stand out, “He’s really good at the whole mafia thing. It’s almost like a secret society.”⁶

In 2014, Graham named Altman president of Y Combinator, and Altman helped turn the firm “into an investment powerhouse.”⁷ He became known as a “startup Yoda” whose connections in Silicon Valley and beyond, combined what some saw an ability to predict the future, made his advice in high demand. Graham explained that when he thought about designing product, he’d wonder what Steve Jobs would do. But for matters of “strategy or ambition” Graham would ask, “What would Sam do?”⁸

Said Graham, Altman was “extremely good at becoming powerful.”⁹ He became the face of OpenAI, testifying to Congress and visiting world leaders. To be clear, he wasn’t writing complex code, said a writer—he was “a dreamer/doer type who’s like a younger version of his cofounder Elon Musk.”¹⁰

From the beginning Altman also had his detractors. At Loopt some executives reportedly urged the board to fire him for his chaotic and, at times, deceptive behavior. As later reported, executives told the board Altman was sometimes untruthful “about matters so insignificant one person described them as paper cuts.” Said Loopt’s former COO, “If he imagines something to be true, it sort of becomes true in his head”—a trait, he suggested, that can be “extraordinary” for ambitious start-up founders but might “make people uncomfortable.”¹¹ (See **Exhibit 2** for OpenAI management as of December 2023).

Background on AI

The quest to develop intelligent machines originated in the 1950s, after mathematician Alan Turing posed the question: “Can machines think?”¹² The “Turing test” as it became known was a thought experiment to test computer intelligence by asking people to guess if they were interacting with a human or machine. In ensuing decades, as computing power increased and costs decreased, the effort advanced, often using games—first chess and later Go—as a proving ground for new developments.¹³

In the 2010s, Google subsidiary DeepMind developed Go-playing program AlphaGo using a “neural network.” Neural networks mimicked the human brain by using computing layers analogous to neurons to process information using weighted variables. Through machine learning, engineers “trained” the neural network, which involved feeding it data to optimize the weights in pursuit of an outcome. DeepMind trained AlphaGo on human expert play. Since the aim was to beat humans, it also trained AlphaGo by having it play against itself. That way, it “learned to discover new strategies.”¹⁴

In 2016, AlphaGo beat Go champion Lee Se-dol. Having conquered games, it seemed time for AI to tackle “real-world problems.”¹⁵ Google had acquired DeepMind in 2014, then a 75-person start-up, for over \$500 million as firms sought to build AI capabilities and hire, by acquisition or otherwise, AI experts. Though DeepMind’s website was “vague” about what it did, a professor said he learned from a co-founder that it was “trying to build a system that thinks.”¹⁶

Promise and perils

“It’s hard to fathom how much human-level AI could benefit society, and it’s equally hard to imagine how much it could damage society if built or used incorrectly,” OpenAI said upon its

founding.¹⁷ Indeed, AI had the promise to improve lives, for example, in healthcare by diagnosing radiological images and in transit by supporting autonomy. Given so many possible applications in those and other industries, as well as productivity gains from automation, robotics, and AI tools for increasing labor efficiency, PwC predicted AI could fuel nearly \$16 trillion in global GDP by 2030.¹⁸

Yet there were issues to overcome. One was bias. In human resources, for example, AI promised to optimize hiring. Yet when Amazon built a “secret AI” system in 2014 to help review candidates, it went haywire when it “did not like women.”¹⁹ Amazon trained it with resumes from the previous decade, a time of “male dominance” in tech, and the system essentially “taught itself that male candidates were preferable.”²⁰ Bias appeared in other critical applications, too. For instance, some facial recognition systems were “falsely identifying African-American and Asian faces,”²¹ and a Black man became “the first known [example] of an American being wrongfully arrested based on a flawed match from a facial recognition algorithm.”²² In finance, worries about AI emerged in areas like mortgage approvals, with a *New York Times* headline asking “Is an Algorithm Less Racist Than a Loan Officer?”²³

Other concerns focused on privacy and data security, such as how to protect the personal data used in training AI systems.²⁴ AI’s potential to replace human workers was another concern, though some argued that with “every disruptive, new technology, we see that the market demand for specific job roles shift.”²⁵ The use of AI in weapons, surveillance, and disinformation raised far-reaching concerns about risks to peace, security, and democracy.²⁶ A researcher cited the risk of powerful AI having a “limited goal of serving only some small group, such as its own creators or those who commissioned it.”²⁷ He questioned whether past innovation could predict the implications of such a giant technological leap.²⁸ Even if challenges with design and function could be addressed, the potential for AI’s misuse by malicious actors was mind-boggling—from using disinformation to topple governments to perpetrating crime and terrorism.

OpenAI Founding

With AI possibly on the cusp of a break-out, Altman worried corporations like Google might make “dangerous AI technologies outside the public eye,” and Tesla CEO Elon Musk expressed “concerns of a dystopian world controlled by powerful AI machines.”²⁹ In December 2015, the two, along with a team of researchers and scientists, announced OpenAI’s founding with Altman and Musk as co-chairs.

In addition to Altman, Musk, Brockman, and others, the founding team included Ilya Sutskever, a “world [expert] in machine learning.”³⁰ Sutskever, considered an “AI superstar,” would “become the soul of the company and its driving force in research.”³¹ As a graduate student in 2012, he had helped design and train a neural network to recognize objects in images. The use of graphics processing units (GPUs) marked a major breakthrough. While powerful GPU hardware was traditionally designed for running intensive video games, it would also prove to be ideal for training neural networks.³²

Non-profit

Musk said he co-founded OpenAI as a non-profit to be “a counterweight to Google” with the “open” referring to “open source”—implying a plan to make code and research public.³³ Musk felt Google wasn’t taking AI safety seriously.³⁴ As a non-profit, OpenAI said it would “advance digital intelligence in the way that is most likely to benefit humanity as a whole, unconstrained by a need to generate financial return.”³⁵ It would be able to “prioritize a good outcome for all over its own self-interest.”³⁶ OpenAI said that Altman, Musk, billionaire Peter Thiel, Y Combinator co-founder Jessica Livingston, LinkedIn co-founder Reid Hoffman and others “committed \$1 billion,”³⁷ though the non-profit ultimately raised only \$130.5 million³⁸ with over \$44 million reportedly from Musk.³⁹ To start, OpenAI

had no CEO, and Brockman and Sutskever were “de facto leaders.”⁴⁰ In an interview, Altman noted that he had little experience with nonprofits, saying, “So I’m just not sure how it’s going to go.”⁴¹

OpenAI’s founding bylaws outlined its governance. Directors alone could elect other directors; set the board size (from one to seven members); remove directors (“at any time ... with or without cause”); and vote on matters like mergers, dissolution, and amending the bylaws. The vote of a majority of directors was “the act of the Board.” The board, which could act without meeting or prior notice if a majority gave written consent, was vested with the authority to exercise and manage all powers and activities of the corporation, either directly or by delegation under the board’s direction. According to OpenAI’s certificate of incorporation, “The specific purpose of this corporation is to provide funding for research, development and distribution of technology related to artificial intelligence. The resulting technology will benefit the public and the corporation will seek to open source technology for the public benefit when applicable. The corporation is not organized for the private gain of any person.”⁴²

OpenAI would say its long-term mission was to ensure the safe development of AGI—“highly autonomous systems that outperform humans at most economically valuable work”—to benefit all of humanity.⁴³ But how it would get there wasn’t clear. In 2016, with Musk worried that Google’s AI efforts could result in “an immortal and superpowerful dictator,” a reporter noted: “It was clear what OpenAI feared,” but not “what it embraced.” When a Google engineer stopped by OpenAI’s offices that May, he told executives that no one really understood OpenAI. Asked the engineer, “what is the goal?” Said Brockman, “Our goal right now ... is to do the best thing there is to do. It’s a little vague.”⁴⁴

For a time, it seemed OpenAI “meandered.” In 2016, OpenAI announced plans to build a robot butler—with third-party robotics powered by its AI—to do housework.⁴⁵ Sometime in 2017, it reportedly “laid off a chunk of its small staff.” Musk, who in 2017 said he was spending just “a couple of percent” of his time on OpenAI,⁴⁶ left in February 2018. Reports cited various reasons for his exit: he was dissatisfied with the progress, wanted more control, disagreed with an idea percolating about starting a for-profit arm, and believed his work at Tesla better positioned him to develop AI there.⁴⁷

OpenAI Charter

In April 2018, OpenAI published the “OpenAI Charter” which outlined four principles to guide execution on the company’s mission. First, OpenAI would ensure that AGI had “broadly distributed benefits” and did not cause harm or give power to a few. “Our primary fiduciary duty is to humanity,” it said. Second was “long-term safety,” with the charter stating, “if a value-aligned, safety-conscious project comes close to building AGI before we do, we commit to stop competing with and start assisting this project.” Third came “technical leadership,” a commitment to “be on the cutting edge” of the capabilities to develop AGI safely. The last principle was “cooperative orientation”—which included a commitment to publish research on “safety, policy, and standards” even if safety concerns precluded publishing the company’s full technical research.⁴⁸

At OpenAI, the charter was “sacred” and viewed as the company’s “backbone.” Written to drive internal alignment, it also set the stage for what was to come. In March 2017, a small team at OpenAI had taken stock of the mission and found that the need for exponential growth in compute power and capital to achieve AGI rendered the non-profit structure untenable, with an observer describing it as “a fatal flaw.” In a nod to this reality, the charter foreshadowed: “We anticipate needing to marshal substantial resources to fulfill our mission, but will always diligently act to minimize conflicts of interest among our employees and stakeholders that could compromise broad benefit.”⁴⁹

A New Direction

Within a few years of its founding, a major breakthrough began to shape OpenAI's strategy. Google published a paper in 2017 about neural networks and language that promised a "hugely optimized ... process of generating coherent text to respond to prompts." The process seemed so promising that OpenAI aligned its workflows around it. Instead of working mostly on individual projects, researchers would now focus "like an engineering organization," said director Adam D'Angelo.⁵⁰

OpenAI built a system it called GPT. Based on the paper's "transformer" architecture, "GPT" stood for "generatively pretrained transformer" — eventually popularized as the term "generative AI." GPT grew more powerful as it was trained on more data. GPT-1 trained on thousands of books and other sources, but GPT-2 trained on the internet. The company's so-called large language models (LLMs) got so good at natural language that the team began to worry about bad actors using them to create spam or disinformation. The risks posed a challenge but also offered evidence OpenAI had struck gold. Said an observer, "that OpenAI was making products smart enough to be deemed dangerous, and was grappling with ways to make them safe, was proof that the company had gotten its mojo working."⁵¹

Meanwhile Altman resigned as Y Combinator's president in March 2019 and became OpenAI's first CEO. Until then, he'd spent around 30 percent of his time on OpenAI. But the company appeared to be at an "inflection point," and Altman worried that it could "disintegrate" without strong leadership.⁵²

Capped-Profit Subsidiary

In March 2019, OpenAI announced the formation of a for-profit entity, which would allow it to raise capital from investors. "We'll need to invest billions of dollars in upcoming years into large-scale cloud compute, attracting and retaining talented people, and building AI supercomputers," it explained.⁵³

The for-profit arm—initially OpenAI LP but eventually known as OpenAI Global LLC⁵⁴—was a "capped-profit" entity meaning investors were limited to a maximum return, set at 100x for early investors and expected to be lowered for any future investors. Excess returns would go to the non-profit. Altman said OpenAI put the cap into place because if AGI came to fruition, it could "capture the light cone of all future value in the universe, and that's for sure not okay for one group of investors to have."⁵⁵ The announcement said that the for-profit's "primary fiduciary obligation is to advance the aims of the OpenAI Charter, and the company is controlled by OpenAI Nonprofit's board."⁵⁶

The non-profit directors, the company added, "must perform their fiduciary duties in furtherance of [the non-profit's] mission—safe AGI that is broadly beneficial."⁵⁷ Investors and employees had to sign agreements acknowledging the charter came first even if it harmed their financial interests. The for-profit now employed most of its staff, and when the company spoke of "OpenAI," it would be referring to the for-profit.⁵⁸ (See **Exhibit 3** for OpenAI's organizational structure as of June 2023).

While OpenAI said raising capital through a for-profit was necessary to develop safe AGI, its stance confused some. "Behold a company that, depending on your time-space point of view, is for-profit and nonprofit. The details are embodied in charts full of boxes and arrows," joked one writer. OpenAI sought to assure the writer; said D'Angelo, "It's my job, along with the rest of the board, to make sure that OpenAI stays true to its mission."⁵⁹ (See **Exhibit 7** for board members as of March 2019.)

With the new structure, OpenAI reportedly raised capital from major VCs including Sequoia, Thrive Capital, and Andreessen Horowitz. Though details weren't entirely clear, one backer, Khosla Ventures,

said it had written a \$50 million check, its largest ever in an initial financing round. Explained Vinod Khosla, “If we lose, we lose 50 million bucks. If we win, we win 5 billion.” In addition, employees received equity-like “profit participation units”⁶⁰ for the first time. Altman, however, decided against owning any himself. He had enough money, he said. He tired of outsiders’ fascination with the decision and joked that he might get one share because then maybe the questions would finally stop.⁶¹

Microsoft Partnership

However much the VCs invested, it wasn’t enough. While GPT-2 had 1.5 billion parameters, GPT-3 would have 175 billion.⁶² Around the time OpenAI was hitting its stride with GPT, Altman met Microsoft CEO Satya Nadella at a conference in July 2018, and Altman’s description of their work caught his attention. In July 2019 they announced an “exclusive computing partnership ... [to] accelerate breakthroughs in AI and power OpenAI’s efforts to create artificial general intelligence (AGI).”⁶³ As part of the companies’ multiyear partnership, Microsoft would invest an initial \$1 billion.⁶⁴

While the deal gave OpenAI access to Microsoft’s deep pockets and vast computing resources, OpenAI had to pony up, too. It agreed to exclusively use Microsoft’s Azure cloud platform for training its AI and give Microsoft an exclusive right to use OpenAI’s technologies in its products. While Altman touted Microsoft as the ideal if not “only” partner, inside OpenAI there was pushback. Some argued it went against the company’s original goal of being a counterweight to large for-profit corporations and worried OpenAI hadn’t yet done enough safety testing for it to enter into a commercial partnership.⁶⁵

Partnership or not, by mid-2019 OpenAI still didn’t have a business model. At a tech event, Altman said the AI might solve that. He said he’d told investors, “Once we build a generally intelligent system ... we will ask it to figure out a way to generate an investment return for you.” The crowd laughed, but it wasn’t a joke. “You can laugh” he said. “But it is what I actually believe is gonna happen.”⁶⁶

The incident hinted at what made OpenAI unique: a fervent belief in AGI. In 2015, such a belief could have been met not just with laughter but ridicule. It was “almost considered a career killer ... to say that you took AGI seriously,” said Altman. A belief that AGI’s arrival would be “one of the greatest moments in human history” seemed all but a prerequisite for working at OpenAI, an observer said. The belief, and optimism about it, may have helped fuel a “moonshot-ish” approach others missed.⁶⁷

Commercial Release

In June 2020, OpenAI released its first commercial product, an application programming interface (API) system that let users integrate its latest model, GPT-3, into their products. By March 2021, more than 300 applications were using it. For example, one company built a system to help companies understand their customer service. A user might type, “What’s frustrating our customers about the checkout experience?” and GPT-3 could quickly integrate data from surveys and help desk and chat logs. It might respond, “Customers are frustrated with the checkout flow because it takes too long to load. They also want a way to edit their address in checkout and save multiple payment methods.”⁶⁸

Because of its limited release, OpenAI was still not well known. That was by design and served two purposes. For one, it reflected OpenAI’s “iterative deployment” approach. In a gradual process including pre-launch safety reviews, starting with just a few users, and reviewing and tackling any issues as they arose, OpenAI sought to methodically address safety.⁶⁹ The other purpose was for AI to appear gradually so that people had “time to adapt to the monumental changes Altman saw coming.”⁷⁰

While OpenAI had typically made much of its code public,⁷¹ with GPT-3 only Microsoft would receive the model's complete code.⁷² OpenAI said the move away from open source reflected its need to commercialize its work to afford "research, safety, and policy efforts;" the model's extraordinary size; and a belief that keeping it under wraps better positioned it to optimize safety.⁷³ While some said OpenAI's iterative approach gave it "a decisive advantage in safety-trained models,"⁷⁴ not everyone agreed. Dario Amodei, who served as OpenAI's safety lead – already worried the Microsoft deal could lead to commercializing GPT before it was safe and undermine commitments OpenAI made in its charter – disagreed with how the API was released. He preferred "a more limited and staged release ... to conduct more testing" on fewer users. He would resign and co-found competitor Anthropic.⁷⁵

Anthropic

Unlike OpenAI, which was founded as a nonprofit, Anthropic was incorporated in Delaware as a public benefit corporation (PBC) in 2021. Delaware law, the company said, "expressly permits the directors of a PBC to balance the financial interests of the stockholders with the public benefit purpose specified" in the certificate of incorporation. Anthropic's purpose was specified as "the responsible development and maintenance of advanced AI for the long-term benefit of humanity." The company believed that the PBC form was compatible with commercial success and stockholder returns and, moreover, that commercial success was necessary to generate the resources needed for Anthropic to achieve its goal of being at the forefront of the most advanced safe AI systems.⁷⁶

Within two years, Anthropic said its PBC model was not enough. Though directors could consider non-shareholders, they were not legally accountable to them. At the close of its series C funding round, Anthropic created the Anthropic Long-Term Benefit Trust (LTBT) governed by an "independent body" of five trustees with no financial interest in the company. Initially chosen by the Anthropic board, trustees for vacant positions would subsequently be selected by the other trustees. Anthropic issued a new share class known as Class T common stock to the trust giving trustees, who served one-year terms, the power – eight months after total investment hit \$6 billion and by May 24, 2027 in any case – to remove Anthropic directors and elect a board majority, three of the five directors at the time.⁷⁷ Preferred holders – typically investors – and common holders – typically employees and founders – would each elect one director.⁷⁸ Unlike most common law trusts, the LTBT was a "purpose trust," run "for the achievement of a purpose rather than the benefit of a particular set of beneficiaries." Its stated purpose mirrored the company's. Anthropic's certificate of incorporation granted holders of Class T common stock the right to receive advance notice of key board actions that might affect the business.⁷⁹

While Anthropic's board would still govern the company, it was accountable to both the LTBT and stockholders, giving it incentives to "balance the public benefit with stockholder interests," said the company. Though a shareholder supermajority could remove a trustee-backed director, trustees would choose the replacement. Anthropic's lawyers said the company and shareholders holding a certain amount of equity for a certain period were authorized to enforce the trust.⁸⁰ Legal observers, however, wondered if shareholders would sue trustees who backed profitable strategies, safe or not.⁸¹ Anthropic called the structure a "corporate governance experiment" and warned against copying it. They would "see how it works."⁸² (See **Exhibit 4** for LTBT trustees and Anthropic board members as of late 2023.)

ChatGPT

By November 2022, as part of its iterative deployment approach, OpenAI planned to launch a public "research preview" of a chatbot using its GPT-3.5 model. While OpenAI said it made "substantial reductions in harmful and untruthful outputs," the system still had some issues. It might sometimes

write “plausible-sounding but incorrect or nonsensical answers,” be “excessively verbose,” respond “to harmful instructions,” or show “biased behavior.”⁸³

After OpenAI launched ChatGPT that month, it gained 1 million users within days; in less than a year, 100 million.⁸⁴ OpenAI approached its success delicately and “discouraged employees from boasting” about it. Though Altman tweeted to announce ChatGPT hit a million users, when Brockman days later tweeted it hit 2 million, Altman had reportedly asked him to delete it out of a sense that “advertising such rapid growth was unwise.”⁸⁵

Still, the growth was tough to hide. It became the “fastest-growing consumer software” ever.⁸⁶ A writer said it became part of his daily life and used it as a tutor; for editing; to write interview questions; to role-play before real-life conversations; and to write computer code. A study showed AI could give programmers a 56 percent productivity boost, though ChatGPT wasn’t designed to code. In training on internet content about coding, it essentially “learned” how, an effect called “emergent behavior.”⁸⁷

Spotlight on AI

With ChatGPT’s record uptake, the world quickly had to navigate new ethical and legal questions. One was concerns about using AI for cheating in school.⁸⁸ Another centered on “hallucinations” — when ChatGPT gave inaccurate information or just seemed to “make things up.” Some worried that such “fabrications” could be dangerous, for example, if a user asked it about legal or health issues. An internal Microsoft memo cautioned that AI chatbots were “built to be persuasive, not truthful,” and “outputs can look very realistic but include statements that aren’t true.”⁸⁹

Still, ChatGPT proved promising in healthcare. By mid-2023, doctors at a top Boston hospital were asking ChatGPT “for possible diagnoses in difficult cases” and training students on using it properly.⁹⁰

ChatGPT also stirred up legal concerns over copyright infringement. In 2023, The New York Times sued OpenAI and Microsoft for an unspecified number of “billions of dollars” for “unlawful copying and use of The Times’s uniquely valuable works.” The companies, it said, were trying to “free-ride” on its product. While OpenAI would dispute the claims,⁹¹ it seemed copyright questions would eventually need to be resolved — one observer predicted “a Supreme Court decision is essentially inevitable.”⁹²

ChatGPT also raised other concerns. Some companies forbade employees from using it over worries they’d input sensitive information⁹³; and it appeared to show bias at times — a study found that when used to write recommendation letters, it might refer to men as an “expert” but women as a “delight.”⁹⁴ Its image generating capabilities also raised questions. In addition to concerns about users bypassing safety filters to create inappropriate pictures,⁹⁵ some worried about peoples’ ability to discern fact from fiction. Said a researcher, “People need to know that the images they see may not be real.”⁹⁶

ChatGPT’s rise brought attention to a movement called “effective altruism.” Loosely based on the utilitarian idea of the greatest good for the greatest number, effective altruism sought to use data and reason to determine what course of action was most likely to maximize benefits to society in any given situation. Many in the effective altruist community argued for slowing AI’s development and emphasizing “safety over speed.” They warned that not doing so could be “apocalyptic” and called for “alignment” — that is, ensuring that AI operated with human values.⁹⁷

OpenAI investor Vinod Khosla saw things a bit differently. He said AI’s potential benefits — “free doctors for everyone,” “tutors for every child on the planet,” and “billions of bipedal robots and agents removing the drudgery of physical labor” — should not be delayed and that technological progress has been the “sole lever by which the human condition has improved.” Said Khosla, “academic talk of

nonsensical existential risk” threatened it all by having “minuscule risks ... hysterically amplified.” And further, geopolitical risk—advancing AI first, before other nations—required full steam ahead.⁹⁸

New Microsoft Investment

ChatGPT was a “money pit” declared a reporter in early 2023. Altman had tweeted that while OpenAI was working to improve costs, ChatGPT’s expense was “probably single-digits cents per chat”⁹⁹—a figure the reporter used to estimate that it was racking up weekly bills in the millions.¹⁰⁰

In January 2023, Microsoft announced a new investment in OpenAI. It already invested another \$2 billion since the initial \$1 billion, and this new investment would bring in \$10 billion more.¹⁰¹ Explained a reporter, “very little” was disclosed about the partnership. Said a Microsoft spokesperson, “All that is publicly shared is that it is a multiyear, multibillion-dollar investment.” Microsoft said the deal wasn’t material and didn’t fall under disclosure rules, a claim some questioned given its focus on AI.¹⁰²

Still, reporters caught wind of a few details. Some reports said the investment wasn’t all cash and included “credits for Azure cloud-computing usage.”¹⁰³ Early reports said Microsoft received a 49% ownership stake, and OpenAI’s website once called Microsoft a “minority owner,” phrasing that later changed to say Microsoft had a “minority economic interest.” Microsoft eventually clarified it “does not own any portion of OpenAI and is simply entitled to share of profit distributions,” which sources said was a 49 percent take, capped by OpenAI’s maximum return. Reporters also learned the deal “excludes any AGI that OpenAI might eventually create.” As OpenAI confirmed on its website,¹⁰⁴ only its board—on which Microsoft did not sit—had the authority to declare when AGI was achieved.¹⁰⁵

Copilot

By 2023, Microsoft was integrating OpenAI technology into numerous products including systems to assist engineers with coding; productivity tools like Excel, PowerPoint, Word, and Outlook; and its Bing search engine. And in November 2023, Microsoft announced Copilot, which it called “your everyday AI companion” designed to “unify” users’ AI experiences across Microsoft’s applications.¹⁰⁶

Amid an “AI frenzy,”¹⁰⁷ seven companies with AI ambitions in one way or another, including Microsoft, GPU maker Nvidia, Google parent Alphabet, Amazon, Apple, Facebook parent Meta, and Tesla more than doubled in aggregate value in 2023 and made up some 28 percent of the S&P 500.¹⁰⁸ ChatGPT, reporters said, set off “a desperate scramble among tech companies,” some of which may have “shunted aside” safety, at least temporarily, as the “instinct to be first or biggest or richest—or all three—took over.” Interest in start-ups—including Anthropic, which raised billions from Amazon and Google—also surged.¹⁰⁹ (See **Exhibit 5** for examples of large AI-related start-up fundraising in 2023)

Regulatory Scrutiny

In 2023, U.S. and UK regulators were eying the Microsoft partnership, and in early 2024, the U.S. Federal Trade Commission ordered Microsoft and OpenAI, as well as Alphabet, Amazon and Anthropic, to provide it with details of their partnerships.¹¹⁰ In late 2023, the EU reached agreement on regulations to help mitigate AI’s risks—for example, certain generative AI would need to show its content was made by an AI—though the rules wouldn’t take effect right away.¹¹¹ Also in 2023, U.S. President Joe Biden issued an executive order to create “new standards for AI safety and security,” including plans for safety testing, consumer privacy protection, preventing discrimination, and developing guidance for “watermarking” content generated by AI.¹¹²

OpenAI in 2023

In February 2023, OpenAI released ChatGPT Plus, a \$19.99 monthly subscription with GPT-4. ChatGPT, still on GPT-3.5, scored in the 10th percentile on the Uniform Bar Exam while GPT-4 scored in the 90th percentile. OpenAI said it was “safer and more aligned,” more accurate, and less likely to respond to content against company policies.¹¹³ (See **Exhibit 6** for an excerpt of OpenAI terms of use).

In March 2023, a group including Elon Musk and Apple co-founder Steve Wozniak signed a letter calling for a six month pause on systems more advanced than GPT-4 to give time to create a safety framework for AI development, though Musk said he knew it wouldn’t be heeded.¹¹⁴ Meanwhile, OpenAI unveiled a “superalignment” team co-led by Sutskever in July 2023 that would seek to answer the question: “How do we ensure AI systems much smarter than humans follow human intent?”¹¹⁵

OpenAI’s commercial efforts, and value, were soaring. Investors had valued OpenAI at close to \$30 billion at the start of 2023, but by September it was reportedly in talks about a secondary transaction—a deal allowing employees to sell profit participation units they already owned—at a valuation as high as \$90 billion. Through the sale planned for early December, employees reportedly “stood to earn millions.”¹¹⁶ OpenAI was expected to earn revenue of \$1 billion in 2023 and “many billions” in 2024.¹¹⁷

Board Evolution

The board’s composition had evolved over the years as both inside and independent directors were added. The board was significantly reshaped in the first half of 2023 by the resignations of three of its then nine members. Reid Hoffman stepped down citing conflicts of interest given his investments in start-ups using OpenAI’s API.¹¹⁸ Shimon Zilis, an executive at Neuralink, a Musk start-up working on computer brain implant technology, also left.¹¹⁹ And that summer, former U.S. Congressman Will Hurd exited to focus on a presidential run.¹²⁰ (See **Exhibit 7** for OpenAI’s board 2015-2023.)

Some aspects of the board’s history were unclear. While the founding bylaws stipulated an upper limit of seven directors, the number was larger at times.¹²¹ And OpenAI’s non-profit disclosures didn’t appear to identify the chair. A 2015 announcement said Musk and Altman were initially co-chairs, and a 2019 announcement indicated Brockman was then chair,¹²² a position he still held in early November 2023.¹²³ While OpenAI once pledged to let anyone view “its governing documents” and “conflict of interest rules,” OpenAI said in late 2023 it was no longer “distributing [such] internal documents.”¹²⁴

With the departures, the board had six members: Altman, Brockman, and Sutskever, and independent directors Tasha McCauley, Adam D’Angelo, and Helen Toner. Toner, appointed in 2021, had done extensive research on AI and served as strategy director for an AI policy research group at Georgetown University. Considered “a critical thinker who isn’t afraid to challenge commonly held beliefs” in AI safety, she’d been involved in effective altruism but said the “community has changed quite a lot since 2014, as have I.”¹²⁵ McCauley, a senior scientist at policy think tank RAND who joined in 2018, also served on the UK board of Effective Ventures, “a federation of organizations dedicated to effective altruism.”¹²⁶ D’Angelo, also a director since 2018, served as Facebook’s CTO before founding question-and-answer website Quora in 2009, where he was CEO. He was said to ask “direct, piercing questions.”¹²⁷ Like Altman, the independent directors had no OpenAI equity.¹²⁸ (See **Exhibit 8** for OpenAI’s board as of November 16, 2023).

Internal Strife

By late 2023, some directors were uneasy about ChatGPT’s growth and questioned how it supported OpenAI’s mission. Toner later said that Altman hadn’t informed the board before ChatGPT’s release—

directors, she said, learned about it on Twitter—and that Altman gave directors “inaccurate information” about OpenAI’s safety processes.¹²⁹ Toner recalled that when the board was considering the release of three somewhat controversial GPT-4 enhancements, Altman said all three were approved by the joint Microsoft and OpenAI safety board. Yet when she asked for proof, she found that only one had been approved.¹³⁰ Members also reportedly felt Altman “was trying to avoid any checks on his power at the company” and had an “unwillingness to entertain any board makeup that wasn’t heavily skewed in his favor.”¹³¹ A report said some directors felt Altman had a “tendency to play different people off one another in order to get his desired outcome” and “tried to ensure information flowed through him.”¹³² McCauley wanted to engage with OpenAI staff, but Altman reportedly said he “needed to be told if the board spoke to employees.”¹³³ Toner also alleged Altman didn’t tell directors about his ownership of OpenAI’s Startup Fund, which invested in AI firms.¹³⁴ OpenAI later said it “put it in Sam’s name” to get it “started quickly,”¹³⁵ calling the structure “temporary” and saying Altman had no financial interest in it.¹³⁶ Some directors felt Altman was not forthcoming about a project to develop AI hardware for which he reportedly met with Middle Eastern investors in September.¹³⁷

Sutskever had reportedly grown “fearful that the technology could wipe out humanity” and “believed that Mr. Altman was bad-mouthing the board to OpenAI executives.” Although board members rarely talked to each other outside of board meetings, Sutskever contacted Toner and urged her to reach out to Murati, who subsequently shared her concerns about Altman’s behavior with the independent directors.¹³⁸ To complicate matters, Sutskever became upset when in October Altman elevated another employee to Sutskever’s same seniority. Some directors saw the fallout from the promotion as “an ultimatum to choose between [Sutskever] and Mr. Altman.” That same month, Toner co-published a paper lauding Anthropic for holding back the release of an AI chatbot as an example of safe AI development. It was in contrast, said the paper, to “the kind of frantic corner-cutting that the release of ChatGPT appeared to spur.”¹³⁹ That didn’t sit well with Altman.¹⁴⁰

Altman told Toner such messaging from a director was troubling, especially as regulators began to set their sights on the industry. In an email to executives, Altman wrote, “I did not feel we’re on the same page on the damage of all this.” Meanwhile, Altman was reportedly advocating for Toner’s removal in an effort that apparently backfired when some directors got the impression he “was trying to pit them against each other.” Reportedly led by Sutskever,¹⁴¹ the board decided to do something.¹⁴²

Altman is Out

At noon on Friday November 17, 2023, Altman logged into a virtual meeting with Sutskever—but Sutskever wasn’t alone. He was joined by the three independent directors. Sutskever reportedly said, “Sam, you’re out. You’re going to be fired.” The board didn’t provide a detailed explanation for their decision, the meeting lasted a few minutes, and Altman found himself locked out of his laptop.¹⁴³

Before the meeting ended, a shocked Altman asked, “How can I help?” They encouraged him to support an interim CEO. In the moment, he agreed. But he soon decided he was going to fight back.¹⁴⁴

Just a few minutes before the board planned to publicly announce Altman’s firing that day, Murati—who the board named as interim CEO—called Microsoft to tell them what was happening. Directors reportedly withheld the information to the very end as they were “nervous about Altman’s ability to use his Silicon Valley influence.”¹⁴⁵ A “shocked” Nadella called Murati to ask for an explanation—but she had none. Nadella then called D’Angelo to ask if a scandal precipitated the firing. D’Angelo confirmed that was not the case but shared little else, and “Nadella remained confused.”¹⁴⁶

The board publicly announced Altman's firing before the stock market closed that day, and Microsoft's stock appeared to fall in its wake.¹⁴⁷ The board said it had conducted "a deliberative review process" that found Altman "was not consistently candid in his communications with the board, hindering its ability to exercise its responsibilities. The board no longer has confidence in his ability to continue leading OpenAI." The statement noted another change: Brockman was off the board but would get to keep his executive role.¹⁴⁸ (See **Exhibit 9** for Microsoft share price 2019 through 2023).

That didn't last long. Later in the day, in a tweet viewed over 20 million times, Brockman said he quit because of Altman's firing.¹⁴⁹ Altman also took to Twitter, at one point writing "i love you all"¹⁵⁰ — some wondered if it might be intentional that the first letters of each word in the phrase spelled "ilya."¹⁵¹ Meanwhile, things at OpenAI were tense. The board got on a call that same day to discuss the situation with executives, and someone reportedly yelled, "This is a coup!" Directors said Altman lied to them but, citing legal concerns, wouldn't say more. When an executive warned that Altman's firing could wreck the company — saying, "It cannot be your duty to allow the company to die" — Toner reportedly replied, "The destruction of the company could be consistent with the board's mission."¹⁵²

In what Toner saw as an "intimidation tactic," a lawyer for OpenAI suggested the board could be in breach of their fiduciary duties if the company collapsed. Toner saw it differently, saying, "OpenAI is a very unusual organization, and the nonprofit mission ... comes first." OpenAI's existence wasn't required to support the non-profit's mission of ensuring AGI benefitted all of humanity, she said.¹⁵³

With Altman and Brockman out, employees and investors stunned, and persisting uncertainty over why Altman was even fired at all, OpenAI seemed to fall into crisis. By November 18, the board was thinking about "bringing Mr. Altman back." D'Angelo and McCauley phoned Altman — they were "rattled" by the call with executives; Google began trying to woo away OpenAI engineers; and customers were looking to move to competitors. Also that day, Murati told the board she wouldn't be interim CEO, and Nadella had a new idea: "What if Mr. Altman joined Microsoft?" That seemed to give Altman some leverage to rejoin "on his terms" or take "OpenAI's talent with him to Microsoft."¹⁵⁴

On a podcast, Nadella said of the situation, "If OpenAI disappeared tomorrow, I don't want any customer of ours to be worried ... because we have all of the rights to continue the innovation."¹⁵⁵

It appeared Altman was on his way to returning as CEO by November 19, but talks with directors broke down. They couldn't agree on who should join the board and "bickered" about Altman being on it. OpenAI named technology entrepreneur Emmett Shear as the new interim CEO. But by the early morning hours on November 20, things had deteriorated even further. Nadella tweeted that Altman, Brockman, and others were joining Microsoft,¹⁵⁶ and almost every OpenAI employee backed a petition calling for the board to resign and threatening to join Altman. Even Sutskever — who had tweeted that he regretted firing Altman and would "do everything I can to reunite the company"¹⁵⁷ — signed it.¹⁵⁸

New Board

By then, "OpenAI's viability was in question" and the board sought to make a deal. At one point during the crisis, the directors had even reportedly approached Anthropic about merging the companies or hiring its CEO — Anthropic's CEO had turned down both proposals.¹⁵⁹ On November 21, directors proposed former Harvard President Larry Summers join the board — they saw him as "an independent thinker" who'd "hold his ground" against Altman¹⁶⁰ — and Altman agreed. Altman would rejoin as CEO — but not as a director — and allow an independent board committee to investigate the crisis; Brockman would return as president and Murati as CTO; and Microsoft would get a non-voting observer seat.¹⁶¹ The directors were out except for D'Angelo, who was in with Summers and, as

chair, Bret Taylor, the former Twitter chair who saw that company through its acquisition by Musk and who the sides had agreed to earlier.¹⁶² (See **Exhibit 1** for OpenAI's board as of November 29, 2023)

Some criticized the new board for its "conspicuously all-male and all-white makeup." Said Taylor, "Of course, Larry, Adam and I strongly believe diversity is essential as we move forward in building the OpenAI board."¹⁶³ But the board, an observer claimed, hinted at diversity problems in AI generally. One analysis from 2018 found that women comprised just 12 percent of machine learning experts.¹⁶⁴

Back together, Altman noted that he and Brockman "are partners in running this company."¹⁶⁵ But for Sutskever, his fate serving in any role at OpenAI seemed unclear. Asked about Sutskever at a Davos interview in January 2024, Altman responded, "no update yet." Surprised, the interviewer sought to clarify: "Is he working at OpenAI?" Said Altman, "I don't know what the exact status of that is."¹⁶⁶

Governance Refresh

The directors had their work cut out for them. Their mandate included nothing less than stabilizing OpenAI, rebuilding the board, and laying a new foundation to govern what some felt could become a historically consequential company. And there was the nagging question of why, over a few chaotic days—with OpenAI otherwise seemingly on top of the world—it almost all came crashing down. By December, Taylor and Summers formed a two-person board committee and hired lawyers from WilmerHale to investigate. Meanwhile, in a process expected to last around six months,¹⁶⁷ it seemed everything was on the table—said Altman, "Clearly the current thing was not good."¹⁶⁸ Did OpenAI need a new governance model? What of the board? Should Altman be on it? Should all three directors stay on? The pace of development clearly wasn't slowing and—depending on who you asked, and their own beliefs—it was possible the board's decisions could reverberate far into the future.¹⁶⁹

Exhibit 1 OpenAI's board of directors, November 29, 2023

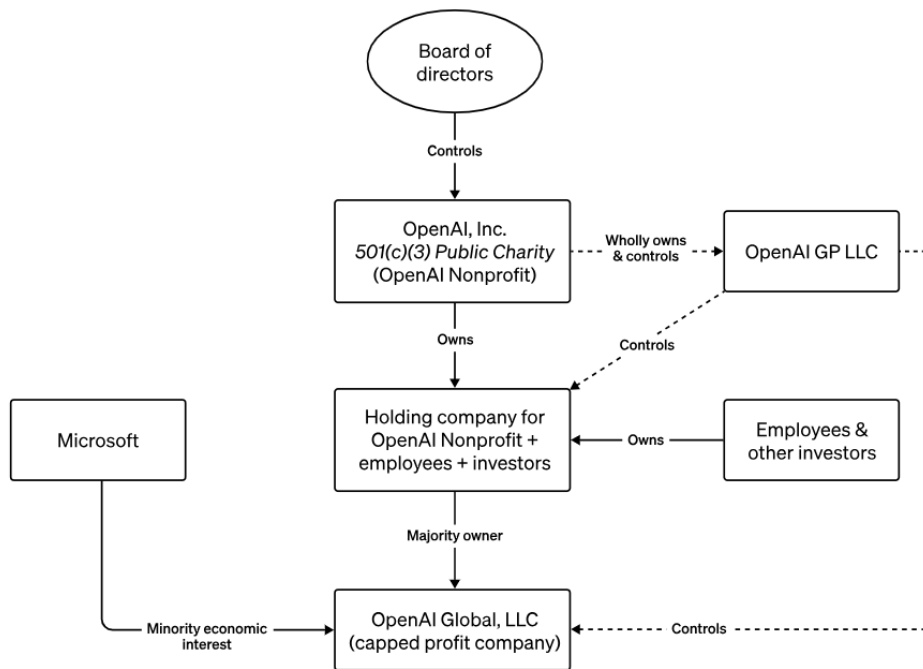
Director	Other Board Positions	Other Experience (Former positions in italics)
Bret Taylor Chairman Independent Director Joined Board: 2023	• Shopify (e-commerce) (NYSE) • Salesforce (NYSE) • Twitter, Chairman (NYSE) • Axon (Taser maker) (Nasdaq)	• Co-founder, Sierra (AI platform for business) • Co-CEO, Salesforce • CEO, Quip (mobile productivity software) • CTO, Facebook • Group Product Manager, Google • Education: Stanford
Larry Summers Independent Director Joined Board: 2023	• Skillsoft (learning tech) (NYSE) • Doma Holdings (real estate tech) (NYSE) • Block (financial tech) (NYSE) • LendingClub (p2p lending) (NYSE)	• Charles W. Eliot University Professor, Harvard • Director & Sr. Economic Advisor, Nat'l Economic Council • President, Harvard • U.S. Secretary of the Treasury • Education: MIT and Harvard
Adam D'Angelo Independent Director Joined Board: 2018	• Asana (collaboration app) (NYSE)	• CEO, Quora (question-and-answer website) • CTO, Facebook • Education: Caltech

Source: Compiled from LinkedIn profiles and data from CapitalIQ, a division of Standard & Poor's, accessed February 2024.

Exhibit 2 OpenAI management, December 2023

Executive	Position	Joined OpenAI	Previous Experience
Sam Altman	CEO (since March 2019)	2015	• President, Y Combinator (start-up incubator) • Founder, Hydrazine (Venture capital firm) • CEO, Loopt (location-based social app) • Education: Stanford (dropped out)
Greg Brockman	President	2015	• CTO, Stripe (payments platform) • Education: MIT (dropped out), Harvard (dropped out)
Mira Murati	CTO	2018	• Product and engineering lead, Leap Motion (augmented reality) • Senior Product Manager, Model X, Tesla • Advanced Concepts Engineer, Zodiac Aerospace (aircraft systems) • Education: Dartmouth
Brad Lightcap	COO	2018	• CFO, OpenAI • Investor, Y Combinator (start-up incubator) • Strategic Finance, Dropbox • Investment Banking Analyst, J.P. Morgan • Education: Duke

Source: Casewriter, compiled from Deepa Seetharaman, et. al., "Sam Altman's Knack for Dodging Bullets – With a Little Help From Bigshot Friends," *The Wall Street Journal*, December 24, 2023, <https://www.wsj.com/tech/ai/sam-altman-openai-protected-by-silicon-valley-friends-f3efcf68>; Greg Brockman, LinkedIn profile, <https://www.linkedin.com/in/thegdb/>; Lollie Barr, Mira Murati profile, CEO Magazine, <https://digitalmag.theceomagazine.com/mira-murati/>; and Brad Lightcap, LinkedIn Profile, <https://www.linkedin.com/in/bradlightcap/>, all accessed February 2024.

Exhibit 3 OpenAI's organizational structure, June 2023

Source: OpenAI, "Our structure," <https://openai.com/our-structure>, June 28, 2023, accessed February 2024.

Exhibit 4a Anthropic board of directors, December 2023

Director	Other Board Positions	Other Experience <i>(Former positions in italics)</i>
Mr. Dario Amodei*	N/A	• CEO and co-founder, Anthropic • <i>VP of Research, Research Director, and Team Lead for AI Safety, OpenAI</i> • <i>Senior Research Scientist, Google</i> • <i>Research Scientist, Baidu (Chinese tech firm)</i> • Education: Stanford and Princeton
Ms. Daniela Amodei*	N/A	• President and co-founder, Anthropic • <i>VP of Safety and Policy, Engineering Manager, and VP of People, OpenAI</i> • <i>Risk Manager, Risk Program Manager, and Lead Technical Recruiter, Stripe (fintech)</i> • Education: University of California, Santa Cruz
Ms. Yasmin Razavi	• Niantic (real-world games; private co)	• General Partner, Spark Capital • <i>Product Manager, Monetization, Snapchat (Snap)</i> • Education: University of Toronto and Harvard
Mr. Luke Muehlhauser	N/A	• Senior Program Officer, Open Philanthropy (philanthropy research) • <i>Exec. Director, Machine Intelligence Research Institute (AI safety research non-profit)</i>

Source: Compiled from Akash Pasricha, “The People With Power at AI Pioneer Anthropic,” The Information, December 13, 2023, <https://www.theinformation.com/articles/the-people-with-power-at-ai-pioneer-anthropic?rc=ztao78>, and biographical data from Pitchbook, LinkedIn and CapitalIQ, a division of Standard & Poor’s, accessed Mar. 2024.

Note: *Dario Amodei and Daniela Amodei are siblings.

Exhibit 4b Anthropic Long-Term Benefit Trust trustees, September 2023

Trustee	Other Board Positions	Other Experience	(Former positions in italics)
Mr. Jason Matheny	• <i>The National Academies of Sciences, Engineering, and Medicine (non-profit)</i> • <i>New Harvest (agriculture science non-profit)</i>	• President & CEO, RAND Corporation (think tank) • <i>Deputy Director for National Security, White House Office of Science and Tech Policy</i> • <i>Coordinator for Technology and National Security, National Security Council</i> • <i>Founding Director, Center for Security and Emerging Technology (AI research)</i> • <i>Commissioner, National Security Commission on Artificial Intelligence</i> • Education: Johns Hopkins	
Ms. Kanika Bahl	• TechnoServe (development non-profit) • Evidence Action (global health non-profit)	• CEO, Evidence Action (global health non-profit) • <i>Managing Director, Results for Development (development non-profit)</i> • <i>Co-chair, UN Commission on Maternal and Child Health Market-shaping Taskforce</i> • <i>EVP, Clinton Health Access Initiative (global health non-profit)</i> • Education: Rice and Stanford	
Mr. Neil Buddy Shah Chair	• Chair, IDinsight (development non-profit)	• CEO, Clinton Health Access Initiative (global health non-profit) • <i>CEO and founding partner, IDinsight (development non-profit)</i> • <i>Managing Director, GiveWell (philanthropy research)</i> • Education: Harvard	
Mr. Paul Christiano	• Ought (AI non-profit)	• Founder and President, Alignment Research Center (AI research) • <i>Led language model alignment team, OpenAI</i> • Education: MIT and UC Berkeley	
Mr. Zach Robinson	N/A	• CEO, The Centre for Effective Altruism (effective altruism charity) • <i>Chief of Staff and Program Officer, Open Philanthropy (philanthropy research)</i> • <i>Director of Product and Strategy, Ivy Research Council (market research firm)</i> • Education: Deep Springs College and Stanford	

Source: Compiled from Anthropic, “The Long-Term Benefit Trust,” September 19, 2023, <https://www.anthropic.com/news/the-long-term-benefit-trust>; Paul Christiano, <https://paulfchristiano.com/>; and biographical and work experience from Pitchbook, LinkedIn and CapitalIQ, a division of Standard & Poor’s, all accessed March 2024.

Exhibit 5 Examples of AI-focused start-ups raising > \$250 million in a financing round in 2023

Company	Co. Description	Financing Date	\$ Raised	Valuation	Known Investors Include
ANTHROPIC HQ: San Francisco	AI safety and research org.; "Claude" chatbot	12/21/2023 (<i>Rumored / in talks</i>)	\$750m	N/A	Menlo Ventures (<i>Rumored / in talks</i>)
 MISTRAL AI HQ: Paris	Provides open source generative AI models	12/5/2023	\$434m	\$2.2b (est.)	Andreessen Horowitz, Lightspeed Venture Partners
 ALEPH ALPHA HQ: Germany	Provides large language models (LLMs) for orgs.	11/6/2023	\$486m	N/A	Innovation Park Artificial Intelligence, Bosch Ventures
ANTHROPIC HQ: San Francisco	AI safety and research org.; "Claude" chatbot	10/27/2023	\$2b	\$25b (est.)	Alphabet (Google)
 metropolis HQ: Los Angeles	AI computer vision for real-world transactions	10/5/2023	\$1.7b	N/A	Eldridge and 3L Capital
ANTHROPIC HQ: San Francisco	AI safety and research org.; "Claude" chatbot	9/25/2023	\$4b	N/A	Amazon
Inflection HQ: Palo Alto	Large language models (LLMs); AI assistant "Pi"	6/29/2023	\$1.3b	\$4b	Microsoft, Reid Hoffman, Bill Gates, Eric Schmidt, and Nvidia
 cohere HQ: Toronto	Large language models (LLMs) for enterprises	6/2/2023	\$270m	\$2b	Inovia Capital and Nvidia
ANTHROPIC HQ: San Francisco	AI safety and research org.; "Claude" chatbot	5/23/2023	\$450m	\$5b	Salesforce Ventures, Spark Capital, Zoom Ventures
 OpenAI HQ: San Francisco	AI tech; large language models (LLMs); ChatGPT	1/23/2023	\$10b	\$29b (Est.)	Microsoft

Source: Casewriter, compiled from Pitchbook and Chris Metinko, "The Biggest Of The Big: AI Startups Raised Huge...", CrunchBase, <https://news.crunchbase.com/ai/biggest-ai-startups-openai-msft-eoy-2023/>, accessed February 2024.

Exhibit 6 Excerpt of OpenAI terms of use, effective January 31, 2024

What You Cannot Do. You may not use our Services for any illegal, harmful, or abusive activity. For example, you may not:

- Use our Services in a way that infringes, misappropriates or violates anyone's rights.
- Modify, copy, lease, sell or distribute any of our Services.
- Attempt to or assist anyone to reverse engineer, decompile or discover the source code or underlying components of our Services, including our models, algorithms, or systems (except to the extent this restriction is prohibited by applicable law).
- Automatically or programmatically extract data or Output (defined below).
- Represent that Output was human-generated when it was not.
- Interfere with or disrupt our Services, including circumvent any rate limits or restrictions or bypass any protective measures or safety mitigations we put on our Services.
- Use Output to develop models that compete with OpenAI.

Source: Excerpted from OpenAI, "Terms of use," Effective Jan. 31, 2024, <https://openai.com/policies/terms-of-use>, accessed February 2024.

Exhibit 7 Evolution of OpenAI's board of directors, 2015-2023 (2023 as of November 16, 2023)

Directors	2015	2016	2017	2018	2019	2020	2021	2022	2023	Example Director Experience
Mr. Elon Musk	B/Ch	B/Ch	B/Ch	B/Ch						CEO, Tesla
Mr. Sam Altman	B/Ch	B/Ch	B/Ch	B/Ch	B	B	B	B	B	CEO, OpenAI
Mr. Chris Clark		B	B							Executive, OpenAI (former COO)
Mr. Holden Karnofsky*			B*	B*	B*	B*	B*			Director AI Strategy, Open Philanthropy
Mr. Greg Brockman			B	B	B/Ch	B/Ch	B/Ch	B/Ch	B/Ch	President, OpenAI
Mr. Ilya Sutskever			B	B	B	B	B	B	B	Chief Scientist, OpenAI
Ms. Sue Yoon*				B*	B*					Director Machine Learning, Google
Mr. Adam D'Angelo*				B*	B*	B*	B*	B*	B*	CEO, Quora
Ms. Tasha McCauley*				B*	B*	B*	B*	B*	B*	Scientist, RAND Corporation
Mr. Reid Hoffman*					B*	B*	B*	B*		Co-Founder, LinkedIn; VC
Ms. Shivon Zilis*						B*	B*	B*		Executive, Neuralink
Mr. Will Hurd*							B*	B*		Former U.S. Congressman
Ms. Helen Toner*							B*	B*	B*	Strategy Director, CSET
* = Independent Director; B = Board Director; Ch = Board Chair										

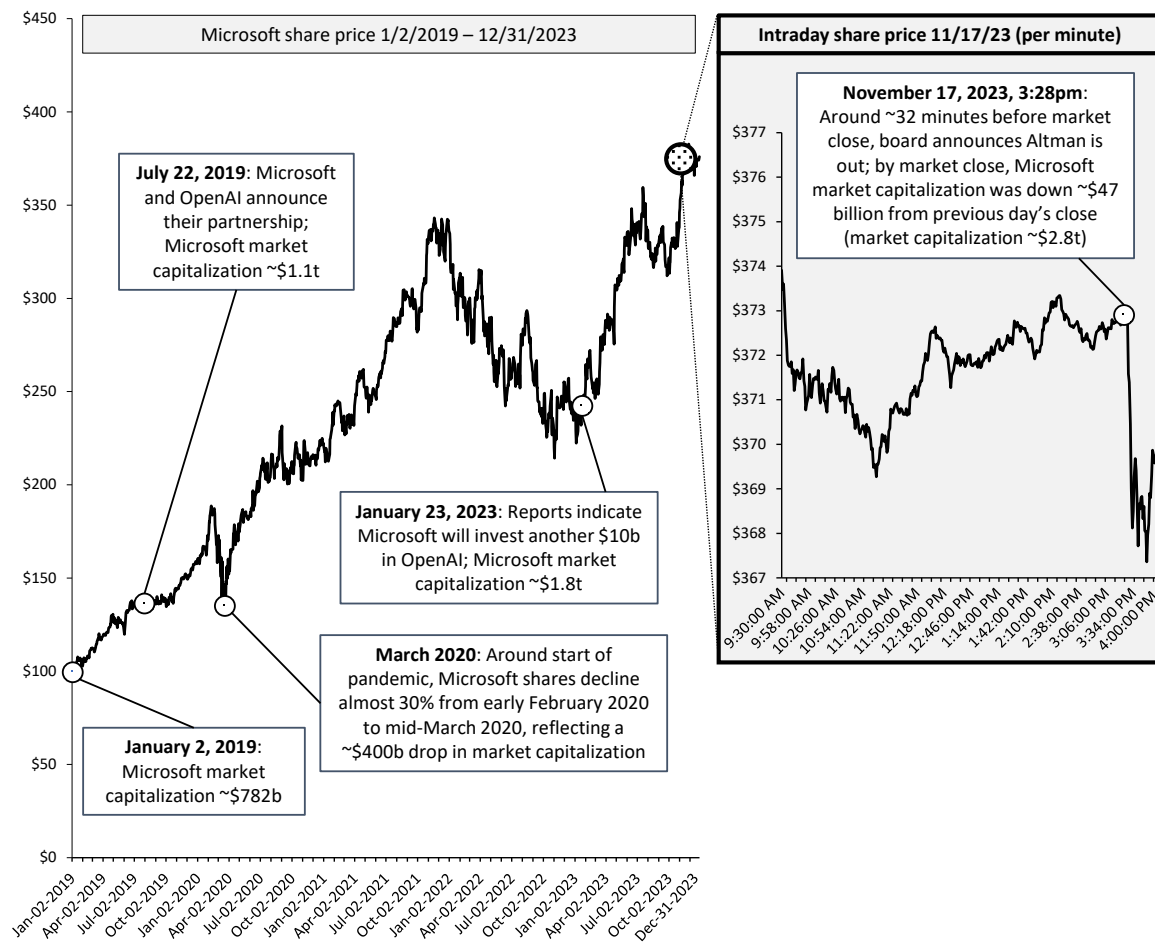
Source: Compiled from media reports 2015, 2023; 2016-2022 from OpenAI Forms 990, GuideStar Inc., accessed February 2024.

Note: Except for 2015 and 2023 (showing board as of 11/16/23), exhibit shows directors serving at any time in given year as reported on Form 990 disclosures. Since the 2016-2022 disclosures show directors serving at any time in a given year – and not on a specific date – some directors listed in a year may not have overlapped in service with all directors listed that year. Non-OpenAI employees and executives assumed to be independent. Assumes Altman and Musk were co-chairs 2015 to 2018, the year Musk left. Based on media reports, Brockman was board chair by 2019. Evolution assumes Brockman consistently served as chair up to 2023, when reports indicated that he was still the chair. OpenAI's 2018-2022 Form 990 disclosures indicate D'Angelo, McCauley, and Toner, the independent directors as of Nov. 16, 2023, received no compensation for board service.

Exhibit 8 OpenAI's board of directors, November 16, 2023

Director	Other Board Positions	Other Experience	(Former positions in italics)
Sam Altman CEO Joined Board: 2015	• <i>Expedia (travel) (Nasdaq)</i> • <i>Green Dot (fintech) (NYSE)</i> • <i>Loopt (social app) (private)</i> • <i>Reddit (social media) (privately-held at time)</i>	• CEO and Co-Founder, OpenAI • <i>President, Y Combinator (start-up incubator)</i> • <i>Founder, Hydrazine (Venture capital firm)</i> • <i>CEO, Loopt (location-based social app)</i> • Education: Stanford (dropped out)	
Greg Brockman Board Chairman President Joined Board: 2017	N/A	• President and Co-Founder, OpenAI • <i>CTO, Stripe (payments platform)</i> • Education: MIT (dropped out), Harvard (dropped out)	
Ilya Sutskever Chief Scientist Joined Board: 2017	N/A	• Chief Scientist and Co-Founder, OpenAI • <i>Research Scientist, Google</i> • Education: University of Toronto and Stanford	
Tasha McCauley Independent Director Joined Board: 2018	• GeoSim Systems (geospatial technologies) (private co)	• Senior Management Scientist, Adjunct, RAND Corporation (think tank) • Director, Ten to the Ninth Plus Foundation (human wellbeing amid tech change) • <i>CEO, GeoSim Systems (geospatial technologies)</i> • Education: Bard College and USC Marshall School of Business	
Adam D'Angelo Independent Director Joined Board: 2018	• Asana (collaboration app) (NYSE)	• CEO, Quora (question-and-answer website) • <i>CTO, Facebook</i> • Education: Caltech	
Helen Toner Independent Director Joined Board: 2021	N/A	• Strategy Dir., Ctr. for Security and Emerging Technology (AI research) • <i>Research Affiliate, Center for the Governance of AI, Oxford (AI research); Sr. Research Analyst, Open Philanthropy Project (philanthropy research); Research Analyst, GiveWell (philanthropy research); and Analyst, Vesparum Capital (AI technology)</i> • Education: Georgetown, Tsinghua University (China), and U. of Melbourne (Australia)	

Source: Casewriter, biographical and work experience data compiled from LinkedIn profiles and data from CapitalIQ, a division of Standard & Poor's, accessed Feb 2024. Altman details from "Sam Altman's Knack for Dodging Bullets—With a Little Help From Bigshot Friends," *The Wall Street Journal*, December 24, 2023, accessed February 2024.

Exhibit 9 Microsoft five-year share price, 2019–2023 and intraday November 17, 2023

Source: Casewriter, Microsoft Corporation (Nasdaq:MSFT) share prices 2019–2023 from Capital IQ, Inc., a division of Standard & Poor's, and Microsoft intraday minute-by-minute share price tick data from FactSet, both accessed February 2024.

Endnotes

- ¹ Jordan Novet, "OpenAI's nonprofit arm showed revenue of \$45,000 last year, even though company is worth billions," CNBC, December 12, 2023, <https://www.cnbc.com/2023/12/12/openai-nonprofit-arm-45000-in-2022-revenue-company-worth-billions.html>, accessed February 2024.
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- ³ Deepa Seetharaman and Berber Jin, "OpenAI Seeks New Valuation of Up to \$90 Billion in Sale of Existing Shares," *The Wall Street Journal*, September 26, 2023, <https://www.wsj.com/tech/ai/openai-seeks-new-valuation-of-up-to-90-billion-in-sale-of-existing-shares-ed6229e0>, accessed February 2024.
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